Tactix: Bounded-Memory Certificate Routing

Report to Andrew

25 September 2026

Outcome

The benchmark proof router and binary-DSR-to-DRAT adapter now process native binary DSR certificates in bounded 1-MiB chunks. Previously they loaded a whole certificate into Python memory, then copied it into a second mutable proof-sized allocation for conversion. This was a general resource risk for large Tier-D proofs under the 32-GiB task limit. The solver, proof emitter, checkers, canonical no-option DSR contract, and all active frozen pins are unchanged. The detailed technical record is AM435.

Validation and limits

The new regression forces a varint across a chunk boundary and checks parsed lines, route selection, exact converted bytes, witnessed-line rejection, malformed-stream rejection, absence of partial output, and no unbounded proof read. Focused oracle/binary/runtime tests pass 46/46. The required fuzz harness self-test passes its 36 unit tests, both reference legs without unexpected failures, and the known-bad controls. Reference-leg timeouts are not verification and this tooling result does not establish N2 or a solver fuzz gate. Campaign-owned Claude/Opus fuzz remains separate; Fable is not used.

The AM427 Sweep Tier-C comparison remains unscorable: 29/30 checked in each arm, with the shared AES proof reaching the original 4,800-second checker limit. The bounded extended-checker array `1723443` remains running on the old, immutable router pin and identical 238,948,969-byte AES proofs. Its 45,000-second checker verdicts are pending, so proof identity is not proof verification. The current-source full Debug job `1723391` also remains running. Neither is altered by AM435.

Next decisions and work

Read the terminal checker stages from both AES diagnostic tasks, then design a matched, fully checked successor Tier-C gate if they verify. Separately, repair the general H2 checker-visible proof-state gap under the canonical no-option DSR contract, or explicitly decide to audit a changed checker contract. N1–N4, the retained cumulative gates, and the live PB rule gap remain open. The learning-readiness review still calls for a shared default-exact inprocessing slot controller and owned, safe-point live branching/full-reduction strategy handles; this routing change does not remove those policy-layer refactors. Submit the requested full 5,000-second Tier-D comparison on `cpu-dense-hwulab-q` only after the goal's correctness and performance gates are satisfied.